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Reflection Paper - Big Data and NoSQL Systems

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I learned that big data involves huge amounts of information that grow very fast and are difficult to manage using traditional databases. The discussion taught me about the 4V's of big data which are volume, velocity, variety, and value. I also learned that NoSQL databases are designed to handle modern data more efficiently because they are flexible and scalable. Technologies like Hadoop, Spark, and Kafka help process and analyze data quickly, especially in real-time systems.

The most interesting part for me was learning about the different types of NoSQL databases such as document, key-value, column, and graph databases. I found it interesting because each type has its own purpose depending on the kind of data being handled. I was also amazed by how companies use big data for fraud detection, customer recommendations, and decision-making.

Some concepts were confusing at first, especially distributed systems and the CAP theorem, because they involve balancing consistency and availability in databases. However, I slowly understood that these are important for handling large-scale applications.

I realized that big data is very different from traditional data because it deals with larger, faster, and more complex information. NoSQL systems are important because they make storing and processing this data easier and faster. After the discussion, I understood how important data is in modern technology and how these concepts can be useful in future careers related to IT and data analytics.